

Abraham Khan

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Ph.D. Candidate in Mathematics

Overview

As a computational mathematician, I develop fast algorithms for densely structured linear and multilinear algebra problems. My current research focuses on developing tensor-based algorithms with an eye towards applications in scientific computing and machine learning; in particular, my focus is on efficiently computing kernel matrices. My broader vision is to develop efficient numerical methods that scale efficiently for larger problem sizes and mitigate the curse of dimensionality. The ultimate goal is for the complexity to grow linearly with the problem's intrinsic dimension and overall size.

Research Interests

Numerical linear algebra; multilinear algebra; machine learning; approximation theory; theoretical computer science.

Education

Ph.D., Mathematics, North Carolina State University *2020–present (expected March 2026)*
Raleigh, NC

Dissertation: Parametric kernel approximations using tensors.

Advisor: Dr. Arvind K. Saibaba.

Qualifying exams: Matrix Theory; Functional Analysis; Abstract Algebra.

B.S., Mathematics, University of South Carolina *2014–2018*
Columbia, SC

B.S., Computer Science, University of South Carolina *2014–2018*
Columbia, SC

Research Experience & Appointments

Research Assistant, North Carolina State University *2022–present*
Raleigh, NC

Advisor: Dr. Arvind K. Saibaba.

NSF RTG Trainee in Randomized Numerical Analysis.

Work supported by NSF grants (DMS-1745654, DMS-1845406, DMS-2411198).

- Developed the *Parametric Tensor Train Kernel (PTTK)* method, a tensor-train based method for computing parametric low-rank approximations of parametric matrices across the full parameter domain.
- Developed *parametric hierarchical matrices*, a new class of hierarchical matrices that efficiently approximates parametric kernel matrices across the full parameter domain.
- Currently investigating hierarchical matrix approximation using tensor decompositions.

Givens Intern, Argonne National Laboratory *Summer 2024*
Lemont, IL

Mentor: Dr. Vishwas Rao.

- Explored data-sparse approximations (hierarchical matrices) and developed the concept of parametric hierarchical matrices.
- Continued this work after the internship to further develop the parametric hierarchical matrix format.

Undergraduate Researcher, University of South Carolina

2016–2018

Columbia, SC

Mentor: Dr. Joshua Cooper.

- Explored a problem that is at the intersection of order theory and theoretical computer science: determine the computational complexity of counting the number of linear extensions of a dimension two semiorder.

Publications

- **Khan, A.**; Saibaba, A. K. (2025). *Parametric kernel low-rank approximations using tensor train decomposition*. *SIAM Journal on Matrix Analysis and Applications*, **46**(2), 1006–1036. (33 pages).
- **Khan, A.**; Chen, C.; Rao, V.; Saibaba, A. K. (2025). *Parametric hierarchical matrix approximations to kernel matrices*. arXiv preprint [arXiv:2511.03109](https://arxiv.org/abs/2511.03109) [math.NA]. (39 pages).
- Scott, M.; **Khan, A.**; Kilmer, M. E.; Hu, X.; Saibaba, A. K. *Hierarchical matrix approximation using tensor decompositions*. Manuscript in preparation.

Software

- **PTTK Method** — [GitHub link](#). Implementation of the PTTK method (Python).
- **Parametric Hierarchical Matrices** — [GitHub link](#). Implementation of the parametric hierarchical matrix approach (Python).
- **Tensor-Train Greedy-Cross** — [GitHub link](#). Implementation of the greedy restricted cross interpolation algorithm for tensor trains (Python).

Conferences & Presentations

ICERM Workshop on Randomized Numerical Linear Algebra

February 2–6, 2026

Providence, RI

Poster: “Parametric Hierarchical Matrix Approximations to Kernel Matrices.”

SIAM Conference on Computational Science and Engineering (CSE25)

March 3–7, 2025

Fort Worth, TX

Talk: “Parametric Kernel Low-Rank Approximations Using Tensor Train Decomposition.”

SIAM Conference on Mathematics of Data Science (MDS24)

October 21–25, 2024

Atlanta, GA

Poster: “Parametric Kernel Low-Rank Approximations Using Tensor Train.”

Workshop on Sparse Tensor Computations

October 18–19, 2023

Chicago, IL

Poster: “Efficient Low-Rank Parametric Kernel Matrix Approximation Using the Tensor Train Decomposition.”

Internal Talks

Randomized Numerical Analysis RTG

November 8, 2024

Raleigh, NC

Talk: “Parametric Kernel Low-Rank Approximations Using Tensor Train.”

Computing Skills

Python · C++ · C · MATLAB · Git · Linux · HPC

Teaching Experience

Teaching Assistant, MA 305: Introductory Linear Algebra and Matrices, North Carolina State University

Fall 2020

Raleigh, NC

- Held office hours; graded assignments and exams.

Teaching Assistant, MA 242: Calculus III, North Carolina State University

2021–2022

Raleigh, NC

Terms: Spring 2021; Fall 2021; Fall 2022.

- Led recitations; held office hours; graded assignments and exams.

Teaching Assistant, MA 121: Elements of Calculus, North Carolina State University

Spring 2022

Raleigh, NC

- Held office hours; graded assignments and exams.